## 1. Talk 1 (15.10.2026, Kasra). Why do formal group laws arise?

**Main idea.** Complex-oriented cohomology theories $E$ have Chern classes, and the tensor product of line bundles turns $E^*(\mathbb{CP}^\infty) \cong E^*[[t]]$ into a formal group law: this is where formal group laws come from in topology.

**Main references:** Lecture 1 [1]; [2] (§§3.3–3.5, 4.3, 4.8, 5.1, 5.4, 9.1–9.2) for characteristic classes, $BU(n)$, complex $K$-theory and complex orientations.

- Define Chern characteristic classes, discuss operations on the set of vector bundles.
- State the existence of the classifying space $BU(n)$ and show a sketch of the computation of its cohomology
- Define complex-oriented and even cohomology theories, explain their importance
- Work out two main examples: complex K-theory and integral cohomology, explain how formal group laws arise in these two examples
- State [Proposition 1.8] ($E^*(\mathbb{CP}^\infty) \cong E^*[[t]]$) and its consequence for $E^*(BU(n))$; construct $F_E(t_1, t_2) = m^*(t)$ from the tensor product map $m\colon BU(1) \times BU(1) \to BU(1)$.
- Define a formal group law

## 2. Talk 2 (22.10.2026, Maqbool). Equivalences of Formal Groups

**Main idea.** A formal group law should be thought of as a group structure on a formal neighbourhood of the origin. We set up the functor-of-points language (étale sheaves, formal groups) that makes this precise, and see that over $\mathbb{Q}$ all formal group laws are isomorphic.

**Main references:** Lectures 2, 3 and 5 [1]; [4], [5] for background.

- Briefly outline the picture from differential geometry: the beginning of Lecture 2 [1] and I.'s notes or any other source.
- Introduce étale sheaves as geometric setting, follow Lecture 2 [1], skip the part about the category of elements: [Definition 2.1], [Remark 2.2], some examples, [Definition 2.16] and below.
- Define formal groups and show how formal group laws produce formal groups, follow Lecture 3 [1]: [Definition 3.7], [Lemma 3.9], [Definition 3.14]
- Work out the example of complex-oriented cohomology theory and $KU$ if time permits: [Example 3.17] and [Remark 3.18] [1], respectively. Show how $\hat{\mathbb{G}}_m$ and $\hat{\mathbb{G}}_a$ arise as formal groups associated to $KU$ and $H\mathbb{Z}$ respectively.
- Define homomorphisms, isomorphisms and strict isomorphisms of formal group laws (cf. [Definition 3.22] [1]).
- Sketch that over $\mathbb{Q}$ all formal group laws are equivalent: [Theorem 5.13] [1] using an intuition from differential geometry and taking some statements on faith (such as [Lemma 5.2]).

## 3. Talk 3 (29.10.2026, Gun). Moduli of Formal Groups

**Main idea.** Formal group laws are classified by the Lazard ring, and formal groups by the moduli stack $\mathcal{M}_{\mathrm{fg}}$.

All references are from [1].

- Representability of functors, show moduli problems (for FG and FGL) representable, and what they are represented by.
- Define the functor $\mathcal{F}\mathrm{gl}$ and prove that it is represented by the Lazard ring [Lemma 6.1, Definition 6.2]. State Lazard's theorem without proof [Theorem 6.8] and explain how the proof over $\mathbb{Q}$ follows from [Theorem 5.13]: [Corollary 6.10], and deduce that formal group laws can always be lifted along surjective homomorphisms [Corollary 6.9].

- Define the moduli stack of formal groups $\mathcal{M}_{\mathrm{fg}}$ following [Definition 12.1]. State and prove [Prop. 12.2, Prop. 12.3]. Define sheaves with values in anima [Definition 8.1]

## 4. Talk 4 (Rescheduled! Wed 04.11.2026, 10:00–12:00, Aila). Complex Orientability & MU is the Universal Complex Oriented Theory

**Main idea.** The complex bordism spectrum $MU$ is the universal complex-oriented theory, and by Quillen's theorem its formal group law is the universal one: $MU_* \cong L$. Thus $\mathbb{S} \to MU$ is the topological shadow of $\operatorname{Spec} L \to \mathcal{M}_{\mathrm{fg}}$.

Main reference [1]; for Thom spectra see also [6, §10].

- Motivation: a complex orientation [Definition 9.1] is a homotopy class of maps $\Sigma^{\infty-2}\mathbb{CP}^\infty \to E$ sending the bottom cell to the unit; we look for a homotopy ring spectrum corepresenting orientations [Remark 9.3]. (Optional) Homotopy ring spectra versus $E_\infty$-rings.
- Heuristic: the pointed symmetric algebra $\widetilde{\mathrm{Sym}}(V)$ and its universal property (the paragraph after [Remark 9.3]); a complex-oriented theory "thinks" that $BU(n)$ is $\mathrm{Sym}^n \mathbb{CP}^\infty$ (the paragraph before [Remark 9.4]).
- Thom spaces classically [Definition 9.5], [Example 9.6], and as colimits of sphere-valued functors [6, Def. 10.1, 10.3, Ex. 10.2, 10.5]. (Optional) Un/straightening; comparison of the two definitions [Remark 9.7]. Thom spectra of virtual bundles.
- Define $MU(n)$ [Notation 9.9] and prove [Lemma 9.8]. Then [Remark 9.11]: $MU(0) \simeq \mathbb{S}$, $MU(1) \simeq \Sigma^{\infty-2}\mathbb{CP}^\infty$, so an orientation is a map $MU(1) \to E$ restricting to the unit. The maps $j_n \colon MU(n-1) \to MU(n)$, [Remark 9.10] and Thom classes.
- Define $MU$ [Definition 9.12] and its homotopy ring structure (the paragraph after [Definition 9.12], [2, Lemma 4.4.4]). State [Theorem 9.14] and prove that every orientation comes from a ring map $MU \to E$, using the Milnor sequence. Examples: $H\mathbb{Z}$ and the Todd genus. (Optional) The $E_\infty$-structure [Remark 9.13] and its universal property [7, Rem. 2.33], [8].
- All ring maps $MU \to E$ via changes of orientation. State [Lemma 11.1] and its reformulation [Remark 11.2].
- State Quillen's theorem [Theorem 9.15 = Theorem 11.8].
- Outlook: $MU_*MU$ and [Theorem 11.10]; the Čech nerve of $\mathbb{S} \to MU$ as the topological shadow of $\operatorname{Spec} L \to \mathcal{M}_{\mathrm{fg}}$ (continued in Talk 5).

## 5. Talk 5 (12.11.2026, Lasse). Formal Groups Classify Certain Cohomology Theories

**Main idea.** Comodules over the Hopf algebroid $(MU_*, MU_*MU)$ are the same as quasi-coherent sheaves on $\mathcal{M}_{\mathrm{fg}}$, so $MU$-homology gives a functor $\mathrm{Sp} \to \mathrm{QCoh}(\mathcal{M}_{\mathrm{fg}})$.

**Main references:** Lecture 12 and parts of Lectures 2, 4, 6 and 10 [1]; [2, §11.5].

- State and prove [Proposition 12.2] and [Proposition 12.3] [1], with the decomposition $G_{\mathrm{inv}} \simeq G^s_{\mathrm{inv}} \rtimes \mathbb{G}_m$ [Definition 6.3].
- Recall [Theorem 11.10] [1] from Talk 4: $(MU_*, MU_*MU)$ presents formal group laws with *strict* isomorphisms, i.e. the quotient $\mathcal{F}\mathrm{gl}//G^s_{\mathrm{inv}}$. Define comodules [Definition 10.12] and quasi-coherent sheaves [Definition 4.2] [1], and state Hovey's theorem [2, Thm. 11.5.1].
- State and sketch [Proposition 12.8], using that even gradings are $\mathbb{G}_m$-actions (Lemma 6.5, Talk 3) to divide out the remaining $\mathbb{G}_m$.
- Hence deduce that we have a functor

$$\mathrm{Sp} \longrightarrow \mathrm{QCoh}(\mathcal{M}_{\mathrm{fg}})$$

following [2, Def. 11.5.2], [2, Thm. 11.5.1] as well.

- Outlook, statement only: the Adams–Novikov spectral sequence for the sphere, $H^s(\mathcal{M}_{\mathrm{fg}}, \omega^t) \Rightarrow \pi_{2t-s}\mathbb{S}$ [Theorem 12.10], [Example 12.12] [1].

## 6. Talk 6 (19.11.2026, Li). Frobenius and heights

**Main idea.** Over $\mathbb{Q}$ there is only one formal group; in characteristic $p$ formal groups are distinguished by their height, detected by the elements $v_n$, and $\mathcal{M}_{\mathrm{fg}}$ acquires the height filtration.

**Main references:** Lecture 13 and Lecture 14 up to Proposition 14.3 [1]; [3, Lecture 12].

- Motivation: over $\mathbb{Q}$-algebras all formal group laws are isomorphic (Talk 2), so what is missing is $\mathcal{M}_{\mathrm{fg}}$ over $\mathbb{F}_p$-algebras.
- Define the Frobenius endomorphism and the relative Frobenius [Definition 13.1] [1]; optionally, maps factoring through it [Proposition 13.7] [1]
- Define what it means to have height at least $n$ [Definition 13.8] [1] + explain its connection with a "usual" definition (see for example [3, Lecture 12, Def. 13], via [Corollary 13.19] below) + [Example 13.16] [1]
- Explain that not all formal groups have a height [Definition 13.11] and below, [Corollary 13.13] [1]; the additive formal group has infinite height [Example 13.14] [1]
- Define the elements $v_n$ [Definition 13.17] [1] + [Corollary 13.19] [1]
- Introduce the height filtration on the moduli stack of formal groups [Proposition 14.1] + [Proposition 14.3] [1] (with the invariant ideal [Definition 14.2] as notation inside Proposition 14.3)

## 7. Talk 7 (26.11.2026, Nikita). How to go from Formal Groups to Cohomology Theories: Landweber Exactness

**Main idea.** Which formal groups come from cohomology theories? Landweber's exact functor theorem answers this in terms of the $v_n$ (flatness over $\mathcal{M}_{\mathrm{fg}}$), and produces $KU$ and many other examples. This ends the first, more algebraic part of the seminar.

**Main references:** Lectures 19 and 20 [1].

- Show how to construct cohomology theories from formal groups: [Construction 19.1] [1], and the first answer, [Corollary 19.4]: flatness over $\mathcal{M}_{\mathrm{fg}}$ (not over $L$) is enough.
- Formulate the Landweber Exact Functor Theorem (LEFT): [Theorem 19.5] [1], in terms of the famous elements $v_n$ from Talk 6, and its geometric form [Theorem 19.11] via the height filtration $\mathcal{M}_{\mathrm{fg}} \supseteq \mathcal{M}_{\mathrm{fg}}^{\geq 1} \supseteq \mathcal{M}_{\mathrm{fg}}^{\geq 2} \supseteq \cdots$, with a sketch of the proof.
- Landweber exact formal group laws [Definition 20.2] and the even periodic theories $E_F$ [Construction 20.4]. Examples: $KU$ and the Conner–Floyd isomorphism [Example 20.5], [Remark 20.6]; periodic complex bordism [Example 20.7] [1].

## 8. Talk 8 (03.12.2026, William). Intro to Ring Spectra and Cohomology Theories $E_\infty$, $E_1$

**Main idea.** The transition to the $\infty$-categorical part: $E_1$- and $E_\infty$-rings, their modules, $MU$ as an $E_\infty$-ring, the Čech nerve of $\mathbb{S} \to MU$ and homotopy fixed points.

**Main references:** [9, §§5.3–5.6]; [6, §7.1, §10.2]; the notes of Talk 1 of the K-theory seminar [10]. References in brackets are from [1], unless cited otherwise.

- Recall symmetric monoidal $\infty$-categories, lax symmetric monoidal functors and CAlg: [9, Def. 5.18, Thm. 5.25, Def. 5.46, Def. 5.49, Def. 5.56, Ex. 5.53–5.55].
- Define cartesian monoids, $E_1$-monoids and monoidal $\infty$-categories following the K-theory seminar, Talk 1, Def. 1.3, Ex. 1.5, Def. 1.6 [10]. Show that symmetric monoidal $\infty$-categories in the sense of [9] are the same as Segal functors $\mathrm{Fin}_* \to \mathrm{Cat}_\infty$ by straightening ([9, Thm. 5.32]; K-theory seminar, Talk 1, Rem. 1.4).

- Define $E_1$-algebras as in [9, Def. 5.49], with $\mathrm{Fin}_*$ replaced by $\Delta^{op}$ (inert maps as in [9, Def. 5.13]), and explain why cartesian monoids do not suffice for ring spectra. Examples: $HR$, $\pi_*$ of an $E_1$-ring [6, §7.1, Warning 7.2]; [9, Ex. 5.53, 5.55, Cor. 5.72.]
- Introduce modules only as notation and a list of facts: [6, §7.1]; $\mathrm{Mod}_{HR} \simeq D(R)$ ([9, Rem. 5.75]).
- Show that $MU$ is an $E_\infty$-ring: [6, Rem. 10.6, Def. 10.8 and the text after it], [Remark 9.13]. This uses the $\infty$-groupoid $\mathrm{Pic}(\mathbb{S})$ of invertible spectra and group completion [10]. Work out $H_*MU$ as a ring: [6, Ex. 10.4, Thm. 10.9, Cor. 10.11].
- Construct the Čech nerve of $\mathbb{S} \to MU$ ([9, §5.6]; [6, §8.1]) and identify its homotopy with the Hopf algebroid of Talks 4–5: [Lemma 11.1], [Remark 11.2], [Theorem 11.10]. Hands-on: $\eta_L$ and $\eta_R$ in degree 2. Optional: Čech nerves of Landweber exact theories (e.g. $KU$) via [Corollary 19.4], [Theorem 19.5].
- Define homotopy fixed points [6, §11.2] and show they are $E_\infty$ for $E_\infty$-rings ([9, Prop. 5.76]). Examples, statements only: $KO_2^\wedge \simeq (KU_2^\wedge)^{h\mathbb{Z}/2}$ [2, §5.8].
- Optional: the $E_\infty$ universal property of $MU$ [11].

## 9. Talk 9 (10.12.2026, Gautham). Spectra are Quasicoherent Sheaves on the Moduli Stack of Formal Groups

**Main idea.** The title is made precise in two ways: spectra are "sheaves on $\mathcal{M}_{\mathrm{fg}}$" in the sense of descent along $\mathbb{S} \to MU$, and the functor $X \mapsto MU_*X$ to $\mathrm{QCoh}(\mathcal{M}_{\mathrm{fg}})$ detects nilpotence.

**Main references:** [6, §8.1, §12.1, §13]; [2, §11.4]; [3, Lecture 25].

References in brackets are from [1], unless cited otherwise.

- Recall the functor $\mathrm{Sp} \to \mathrm{QCoh}(\mathcal{M}_{\mathrm{fg}})$ of Talk 5 ([2, Def. 11.5.2], [Proposition 12.8]) and the Čech nerve of $\mathbb{S} \to MU$ from Talk 8, whose homotopy is the cogroupoid $(L, L[b_1, b_2, \dots])$ presenting $\mathcal{M}_{\mathrm{fg}}$ [Theorem 11.10]. Explain the algebraic model: faithfully flat descent of vector spaces along $\mathbb{R} \to \mathbb{C}$ [6, §8.1].
- Descent of modules: define it [6, Def. 8.2], state [6, Prop. 8.1] and prove [6, Thm. 8.3(1),(2)]; apply it to $\mathbb{S} \to MU$.
- Descent of categories: state [6, Thm. 12.3(1), Cor. 12.4] with a sketch, and [6, Thm. 12.10(1)] (with [6, Def. 12.5] as notation): $\mathrm{Sp} \simeq \lim_\Delta \mathrm{Mod}_{MU^{\otimes \bullet+1}}$. Compare with $\mathrm{QCoh}(\mathcal{M}_{\mathrm{fg}})$ as descent data along $\mathrm{Spec}\, L \to \mathcal{M}_{\mathrm{fg}}$.
- Nilpotence: state the smash form [6, Thm. 13.1] and deduce the ring form [6, Thm. 13.3], [3, Lecture 25, Thm. 3], [2, Thm. 11.4.3]. Prove Nishida's theorem [2, Thm. 11.4.2], [3, Lecture 25, Cor. 4]: what nilpotence tells us about the functor $X \mapsto MU_*X$ is that it loses information only "up to nilpotence".

## 10. Talk 10 (17.12.2026, Sayan). Spectral Primes: Morava E-theory and Morava K-theory

**Main idea.** An overview talk: $\mathcal{M}_{\mathrm{fg}}$ has one geometric point of each height $n$; Morava $K$-theory $K(n)$ is the "residue field" at that point and Morava $E$-theory $E_n$ its "formal neighbourhood", acted on by the Morava stabilizer group. Height 1 ($KU_p^\wedge$) serves as the running example.

**Main references:** [2, §1.5, §1.6, §10.6, §11.2] (the overview); Lectures 14–16, 20 [1]; [3, Lectures 22, 24].

References in brackets are from [1], unless cited otherwise.

- The points of $\mathcal{M}_{\mathrm{fg}}$: recall the height filtration (Talk 6, [Proposition 14.1], [Proposition 14.3]); the Honda formal group law, explicitly via its logarithm [2, §1.5, Thm. 10.6.2], [Example 14.16]; Lazard's uniqueness [Corollary 15.4]: one geometric point of each height. Clue: $\hat{\mathbb{G}}_m$ at height 1.

- Morava $K$-theory as the spectrum "sitting at" that point [2, §1.5], [3, Lecture 22, Def. 5] (black box); why it is a field; $K(n)^*(BC_p)$ [2, Rem. 11.2.5]. Why we care: type of finite spectra, thick subcategories and the periodicity theorem [2, Thm. 1.5.1, Thm. 1.5.2] ("spectral primes"). Clue: $K(1) \subseteq KU/p$.
- Symmetries of the point: the (extended) Morava stabilizer group [Definition 16.10], [Remark 16.12], [2, §1.6]. Clue: $\mathbb{G}_1 = \mathbb{Z}_p^\times$.
- Morava $E$-theory as the spectrum of the "formal neighbourhood" of the point: $\pi_* E_n$ and Landweber exactness [2, §1.6, Def. 14.1.7–14.1.8], [Definition 20.22]; Goerss–Hopkins–Miller [2, Thm. 1.6.1], [Theorem 20.23]. Clue: $E_1 = KU_p^\wedge$ with Adams operations.

## 11. Talk 11 (07.01.2027, Thorger). Forms of K-theory: Getting to Know the First Elliptic Cohomology

**Main idea.** Everything at height 1, by hand: $E_1 = KU_p^\wedge$, its Adams operations and the $K(1)$-local sphere.

***Supposed to be covered:*** *Hands on $E(1) = KU_p^\wedge$. Description of $L_{K(1)}S$. Illustrate on this example the properties covered in the previous talks.*

## 12. Talk 12 (14.01.2027, Priyanka). Chromatic Fracture Square

**Main idea.** A spectrum can be reassembled from its chromatic layers: the chromatic fracture square glues $L_{K(n)}$ and $L_{n-1}$ to recover $L_n$.

***Supposed to be covered:*** *The Chromatic Fracture Square. Bousfield localisation, $L_{E(n)}$ is a smashing localisation. Height filtration.*

## 13. Talk 13 (21.01.2027, Phil). E-theory is a Galois cover of the $K(h)$-Local Sphere

**Main idea.** At height $n$, Morava $E$-theory is a Galois extension of the $K(n)$-local sphere with Galois group the Morava stabilizer group: $L_{K(n)}\mathbb{S} \simeq E_n^{h\mathbb{G}_n}$ (Devinatz–Hopkins).

***Supposed to be covered:*** *The Devinatz–Hopkins Theorem*

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